AgriSathi: Unified AI Farming Platform — Hackathon Plan

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Vision and pitch

AgriSathi is one Bangla-first app that stays with a smallholder farmer from planting to sale. The five features are not five apps. They are five stages of the same season, sharing one farmer profile and one field record.

One-line pitch: "Your AI agronomist, doctor and market agent, in Bangla, in your pocket."

Problem (Bangladesh):

- Most farmers work under 1 hectare and rarely see a government extension officer.
- Pests and diseases such as rice blast, brown planthopper and bacterial leaf blight are often noticed too late.
- Middlemen set farm-gate prices, so farmers sell low at harvest.
- Floods, salinity in the south-west and erratic rain make old planting calendars unreliable.

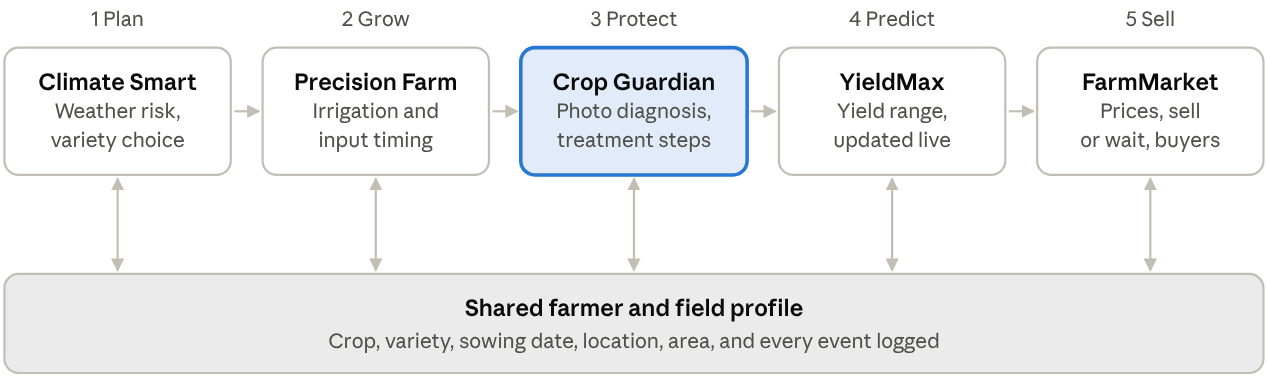
Who it serves: a rice and vegetable farmer with a basic Android phone, patchy data and limited reading in English.

Hackathon strategy: build one connected core loop that really works, and make the other features thinner but visibly linked to it. Judges reward a working flow over five half-built screens.

How the five features connect

The features follow the farmer's season in order, and all of them share one profile. That shared data is what makes this one product instead of five demos.

Five features, one season, one shared profile



Each feature reads the profile and writes events back; Crop Guardian (highlighted) is the hero demo.

AgriSathi season flow · 5 features, 1 shared profile

Example: a Crop Guardian diagnosis is saved to the field, so YieldMax lowers the expected yield and FarmMarket adjusts how much the farmer can list for sale.

MVP scope per feature

Crop Guardian is the hero feature and gets the deepest build. The other four are real but narrow, each fed by the shared field profile.

Feature	Must-have for demo	Stretch if time allows	Depth
Crop Guardian	Photo of a leaf gives disease name, confidence and treatment steps in Bangla	Outbreak map from all reports; voice output	Hero, about 40% of effort
Precision Farm	Daily advice card: irrigate or not, fertilizer timing, based on crop stage and weather forecast	Soil test input; SMS alerts	About 20%
Climate Smart Farming	Weather-risk alerts (heavy rain, heat, flood) with one practical action each; suggest a stress-tolerant variety	District salinity and flood risk layer	About 15%

Feature	Must-have for demo	Stretch if time allows	Depth
YieldMax	Expected yield range for the field, which rises or falls with disease reports and advice followed	Season-over-season comparison	About 10%
FarmMarket	Today's prices in nearby markets, a sell-now-or-wait hint, and a list buyers can see	Buyer matching and in-app offers	About 15%

Cross-cutting must-haves: Bangla interface, one farmer profile (crop, district, field size, sowing date), and a home screen that shows all five as one dashboard.

Architecture and tech stack

Use one backend, one database and one mobile-friendly frontend. Every feature reads and writes the same farmer and field records, which is what makes the five feel like one product.

Layer	Choice	Why
Frontend	React + Vite as a PWA (or Flutter if the team knows it)	Installs on Android, works with weak data, one codebase
Backend	FastAPI (Python)	Same language as the ML code; quick to build REST endpoints
Database	Supabase (Postgres + auth + storage)	Farmer profiles, field records, photos and market listings in one place
Disease model	Fine-tuned MobileNetV3 or EfficientNet-B0 on rice and vegetable leaf images	Small, fast, can later run on-device with TensorFlow Lite
Advice engine	An LLM API, given the farmer profile, weather and diagnosis as context	Turns raw outputs into short Bangla instructions
Rules layer	Plain Python rules for irrigation, fertilizer timing and weather alerts	Predictable and easy to explain to judges

Layer	Choice	Why
Yield estimate	Simple regression or rule-based range from crop, area, variety and stress events	Honest about limited data; good enough to demo
Voice (stretch)	Bangla text-to-speech	Helps farmers with low literacy
Hosting	Render or Railway for the API, Vercel for the PWA	Free tiers, deploy in minutes

Shared data model: Farmer → Field (crop, variety, sowing date, location, area) → Events (diagnoses, advice given, weather alerts, sales). YieldMax and FarmMarket read these events; Crop Guardian, Precision Farm and Climate Smart write them.

Data sources and APIs

All of these are free or public. Check each one on day zero, before the clock starts, so no feature is blocked by missing data.

Need	Source	Used by
Leaf disease images	PlantVillage dataset; rice leaf disease datasets on Kaggle	Crop Guardian
Weather forecast	Open-Meteo API (free, no key)	Precision Farm, Climate Smart
Crop calendars and varieties	BRRI and BARI published guides (e.g. flood-tolerant and salt-tolerant rice varieties)	Precision Farm, Climate Smart
Market prices	Department of Agricultural Marketing (DAM) daily price data; seed a CSV if no API	FarmMarket
Yield baselines	Bangladesh Bureau of Statistics yield-per-hectare figures by crop	YieldMax
Treatment advice	DAE and BRRI pest management guidance, written into a small knowledge file	Crop Guardian

Fallback rule: if a live source fails, use a realistic seeded file. Say so openly in the pitch; judges accept mock data when the pipeline is real.

Team roles

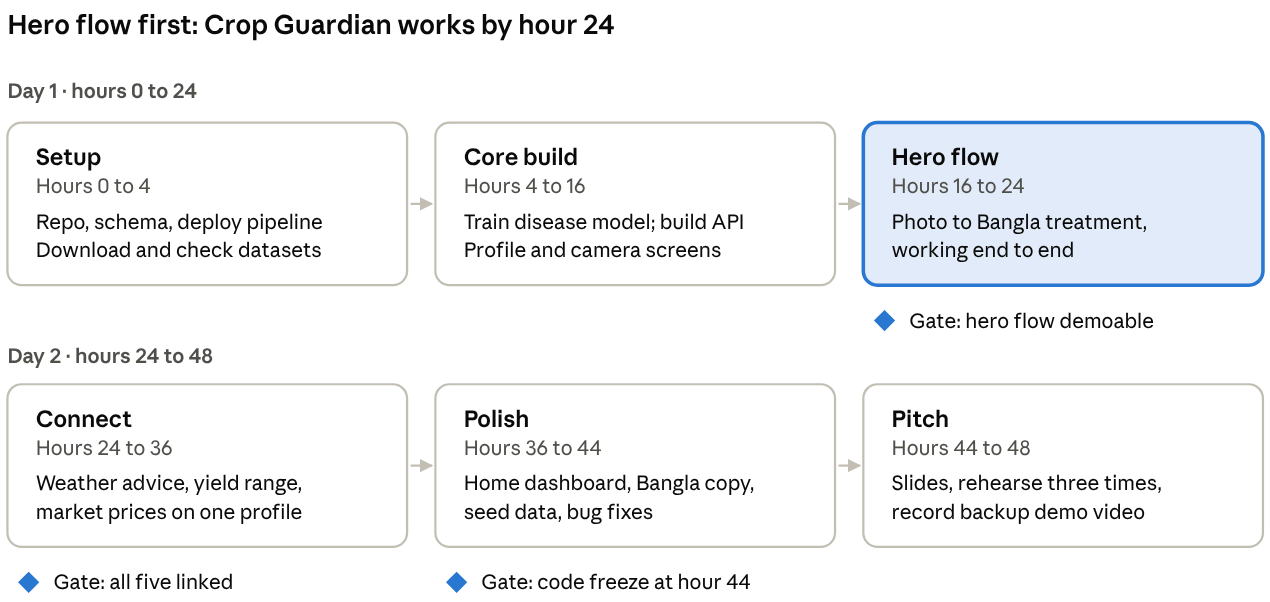
This plan assumes a team of four. With three, the frontend lead also owns the pitch; with five, add a second backend developer for FarmMarket.

Role	Owns	Features
ML lead	Train and serve the disease model; yield estimate logic	Crop Guardian, YieldMax
Backend lead	API, database, weather and price integrations, rules engine, LLM advice prompts	Precision Farm, Climate Smart, FarmMarket
Frontend lead	PWA screens, Bangla UI, home dashboard, camera upload	All five screens

Role	Owns	Features
Product and pitch lead	Farmer persona, data seeding, Bangla copy, testing, slides, demo video backup	Story and demo

48-hour timeline

Get the Crop Guardian flow working end to end by hour 24, then spend day two connecting the other four features to it. If your hackathon is 24 or 36 hours, keep the same order and shrink each phase.



48-hour plan · 6 phases, 3 gates

If the hero-flow gate is missed at hour 24, pause other work until it passes; the cut order under Risks decides what to drop.

Demo script and judging pitch

Tell it as one farmer's season in about 4 minutes. Every step shows one feature handing data to the next.

1. **Meet Rahim (30 s).** A rice farmer in Rangpur with 0.8 hectares. Show his profile being created in Bangla.

2. **Morning advice (40 s, Precision Farm + Climate Smart).** The home screen says to skip irrigation because rain is coming, and warns of a heavy-rain risk with one action to take.
3. **Something is wrong (60 s, Crop Guardian).** A judge photographs a sample diseased leaf. The app names the disease, shows confidence and gives treatment steps in Bangla.
4. **The impact (30 s, YieldMax).** The yield estimate drops, then partly recovers once treatment is logged.
5. **Harvest and sale (40 s, FarmMarket).** Nearby market prices appear with a hint to wait a few days for a better price.
6. **Close (30 s).** One app, one profile, the whole season. State next steps: offline model, voice, a pilot with one farmer group.

Pitch structure for slides: problem, Rahim's story, live demo, how the five connect (use the diagram above), tech, impact numbers, roadmap.

Bring: printed photos of diseased leaves, a backup screen recording, and the app pre-loaded on two phones.

Risks and fallbacks

The biggest risk is scope: five features in 48 hours. The cut order below decides in advance what to drop, so nobody argues at 3 a.m.

Risk	Fallback
Model accuracy is low on real photos	Limit to 4 to 6 common rice diseases; use a vision LLM as backup classifier
Weather or price API is down at demo time	Cache the last response; switch to seeded data
Running out of time	Cut in this order: FarmMarket buyer listing, YieldMax history, Climate Smart risk map, voice. Never cut the Crop Guardian flow or the shared profile
Bangla text looks broken	Use a tested Bangla font (Hind Siliguri or Noto Sans Bengali) from hour one
Live demo fails	Play the backup screen recording and keep talking

Risk	Fallback
Judges ask "is the advice safe?"	Show that advice comes from BRRI and DAE guidance, and that the app tells farmers to confirm chemical use with a local agriculture officer